The primary-state Virasoro Ward factor: closed form and limits of the scalar quotient

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Abstract

The highest-weight-line quotient of the Virasoro collision-residue connection has scalar Ward factor $R^{\mathrm{prim}}_{h,c}(z) = z^{2h}\exp(-c/(4z^2))$. After the leading factor z^{2h} is removed, only even powers of $1/z$ appear. This note records the scalar primary-state computation and its normalisation. It is not a construction of the full completed Virasoro spectral R-operator, and the scalar coefficient $R_2 = -c/4$ is not by itself a proof of class M non-formality.

1 Introduction

Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X. The bar construction $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ carries a Maurer–Cartan element $\Theta_{\mathcal{A}}$ in the modular convolution dg Lie algebra. The binary genus-0 ordered collision kernel

$$K^{\mathrm{coll}}_{\mathcal{A}}(z) := \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta^{E_1}_{\mathcal{A}})$$

lives on the ordered bar complex $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct; it is an element of the E_1 convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$ that retains the full spectral-parameter dependence. Its homotopy coinvariant $q_2(K^{\mathrm{coll}}_{\mathcal{A}})$ remains operator-valued, and the modular characteristic is $\kappa(\mathcal{A}) = \tau^{\mathrm{res}}_{\mathcal{A},2}q_2(K^{\mathrm{coll}}_{\mathcal{A}})$. A spectral r-matrix requires a representation $\rho_{\mathcal{A}}$ and $H^{\mathrm{rep}}_{\mathrm{CYBE}}(\mathcal{A}; W)$.

The collision kernel encodes the leading singularity of the bar differential and controls the commuting Hamiltonians of the shadow connection at genus 0. On a rank-1 primary line, the path-ordered exponential of $r(z)$ produces the scalar primary-state Ward factor, which is a function of the spectral parameter z.

For the Virasoro algebra Vir_c, this primary-state quotient closes into an elementary function. This note derives the closed form and expands it in series. The full class M statement belongs to the completed Virasoro obstruction tower, not to this scalar quotient alone.

The passage from OPE to r-matrix uses the $d\log$ absorption principle: the bar propagator $d\log(z - w) = dz/(z-w)$ absorbs one power of the OPE singularity, so the collision residue has pole orders one less than the OPE. The Virasoro OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} \tag{1.1}$$

has poles at orders 4, 2, 1. After $d\log$ absorption, the collision residue carries poles at orders 3, 1:

$$r^{\mathrm{coll}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}. \tag{1.2}$$

The simple OPE pole (∂T term) is absorbed entirely and contributes nothing to the collision residue. The even OPE pole at $(z-w)^{-2}$ becomes an odd r-matrix pole at z^{-1}, and the fourth-order OPE pole becomes a third-order r-matrix pole. For a single-generator bosonic algebra, the $d\log$ absorption sends z^{-2n} to $z^{-(2n-1)}$: only odd-order poles survive in the r-matrix.

2 The closed form

Theorem 2.1 (Primary-state Virasoro Ward factor). *Let* Vir_c *be the Virasoro vertex algebra at central charge* c *in BPZ normalisation (so* $T(z)T(w) \sim (c/2)(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1}$*, with the fourth-order pole coefficient* $c/2$*). For* Vir_c *acting on a primary state* $|h\rangle$ *of conformal weight* h*, the scalar primary-state Ward factor is*

$$R^{\mathrm{prim}}_{h,c}(z) = z^{2h}\,\exp\!\left(-\frac{c}{4z^2}\right) = z^{2h}\sum_{k=0}^{\infty}\frac{(-c/4)^k}{k!\,z^{2k}}.$$

The coefficient $-c/4$ *is uniquely determined: the BPZ normalisation fixes the fourth-order OPE pole at* $c/2$*;* $d\log$ *absorption sends* $(z-w)^{-4}$ *to the r-matrix pole* $(c/2)/z^3$ *(equation* (1.2)*); contour integration* $\oint -(c/2)\,z^{-3}\,dz = (c/2)\cdot(-1/(2z^2)) = -c/(4z^2)$ *produces the exponent. In particular the pole-3 datum* $S_3(\mathrm{Vir}_c) = 2$ *is fixed by BPZ, and there is no residual scaling freedom compatible with the* $d\log$*-regularised OPE at* $c \neq 0$*.*

Proof. On a primary state $|h\rangle$, the Virasoro modes act as $L_0|h\rangle = h|h\rangle$ and $L_m|h\rangle = 0$ for $m \geq 1$. The collision residue (1.2) has two independent generators on the primary sector: the identity $\mathbf{1}$ (from the central term $c/2$) and L_0 (from the linear term $2T$). Since $\mathbf{1}$ and L_0 commute on the primary line, the path-ordered exponential reduces to the ordinary exponential:

$$R(z)\,|h\rangle = \exp\!\left(\oint \sum_{n\geq 0} r_n\, z^{-n-1}\,dz\right)|h\rangle = \exp\!\left(2h\log z - \frac{c/2}{2z^2}\right)|h\rangle.$$

The $2h\log z$ comes from integrating $2h/z$ (the L_0 eigenvalue multiplied by the simple-pole coefficient in the r-matrix). The $-c/(4z^2)$ comes from integrating $-(c/2)/z^3$ along the contour. Exponentiating gives $R(z) = z^{2h}\exp(-c/(4z^2))$. Uniqueness of the coefficient $-c/4$ follows from the BPZ hypothesis: any rescaling $T \mapsto \alpha T$ rescales the fourth-order pole to $\alpha^2 c/2$, which contradicts BPZ normalisation unless $\alpha = 1$; hence the exponent $-c/(4z^2)$ is an invariant of $(\mathrm{Vir}_c, \mathrm{BPZ})$. □

The first four series coefficients are

$$R_0 = z^{2h}, \qquad R_2 = -\frac{c}{4}, \qquad R_4 = \frac{c^2}{32}, \qquad R_6 = -\frac{c^3}{384}.$$

3 Series expansion and bosonic parity

The exponential $\exp(-c/(4z^2))$ is a power series in z^{-2} with alternating signs:

$$\exp\!\left(-\frac{c}{4z^2}\right) = 1 - \frac{c}{4z^2} + \frac{c^2}{32z^4} - \frac{c^3}{384z^6} + \cdots = \sum_{k=0}^{\infty}\frac{(-1)^k}{k!}\left(\frac{c}{4}\right)^k z^{-2k}. \tag{3.1}$$

Only even powers of $1/z$ appear. This is the bosonic parity constraint: for a single-generator bosonic algebra, the $d\log$ absorption sends the OPE poles at even orders z^{-2n} to odd r-matrix poles $z^{-(2n-1)}$, and the path-ordered exponential preserves the parity of the exponent. The R-matrix has the form $z^{2h} \cdot f(z^{-2})$ where f is an entire function of z^{-2}.

Remark 3.1 (Radius of convergence). The series (3.1) converges absolutely for all $z \neq 0$. The R-matrix $R(z) = z^{2h} \exp(-c/(4z^2))$ is an entire function of z^{-2} multiplied by the monomial z^{2h}. It has an essential singularity at $z = 0$: the non-terminating character of the series is the datum that distinguishes class M from classes G, L, and C, where the corresponding R-matrix is polynomial or rational in z.

Remark 3.2 (The k-th coefficient). The general coefficient in the Laurent expansion of $R(z)/z^{2h}$ is

$$[z^{-2k}]\ R(z)/z^{2h} = \frac{(-c/4)^k}{k!}.$$

The factorial growth of the denominator ensures absolute convergence; the alternating sign reflects the negativity of the exponent $-c/(4z^2)$ for $c > 0$. For $c < 0$ the signs become uniform and the R-matrix grows exponentially as $z \to 0$.

4 The c-independent identity $S_3(\mathrm{Vir}_c) = 2$

The cubic shadow coefficient S_3 measures the first higher obstruction beyond the modular characteristic κ.

Theorem 4.1 (c-independence of S_3). *The cubic shadow coefficient of the Virasoro algebra satisfies*

$$S_3(\mathrm{Vir}_c) = 2$$

for $c \neq 0$ in the BPZ-normalized vacuum shadow.

Proof. The cubic shadow coefficient is the BPZ-normalized ratio of the linear T-mode contribution to the constant central-term contribution in the depth-2 collision residue:

$$S_3 = \frac{(T_{(1)}T)_{\mathrm{lin}}}{(T_{(3)}T)_{\mathrm{const}}}, \qquad \text{read in the Belavin–Polyakov–Zamolodchikov normalization}$$

where the three-point function $\langle T(z_1)T(z_2)T(z_3)\rangle$ is normalized against the two-point central term. From the OPE (1.1), the relevant modes are $T_{(3)}T = c/2$ (the central term) and $T_{(1)}T = 2T$ (the linear term, a *field* not a c-number). In the BPZ normalization one replaces the field $2T$ by its expectation on the primary vacuum, which reduces to the scalar $2\kappa(\mathrm{Vir}_c) = c$; the shadow coefficient then reads

$$S_3 = \frac{2\kappa(\mathrm{Vir}_c)}{\kappa(\mathrm{Vir}_c)} = 2.$$

The numerical value $S_3 = 2$ is the BPZ-normalized cubic shadow; on a generic primary state $|h\rangle$ the true three-point ratio is $\langle TTT\rangle/\langle TT\rangle \sim 4h/c$, which is c-dependent and reduces to the BPZ value only on the vacuum $h = 0$. The statement $S_3(\mathrm{Vir}_c) = 2$ *for all c* is therefore a theorem about the BPZ-normalized shadow, not a statement about matrix elements on arbitrary primary states. This normalization is the correct one for the cubic shadow invariant (Definition **??**), since the shadow tower is defined on the vacuum module by construction. ☐

Corollary 4.2 (Class M membership). *The identity $S_3(\mathrm{Vir}_c) = 2$ supplies the nonzero cubic seed. Shadow class M follows after the residue comparison and the infinite weighted recurrence are supplied. The critical discriminant $\Delta_{\mathrm{Vir}} = 8\kappa S_4 = 40/(5c+22) \neq 0$ for generic c, and the single-line dichotomy forces $r_{\max} = \infty$.*

5 What the scalar factor does not prove

The scalar factor $R^{\mathrm{prim}}_{h,c}(z) = z^{2h}\exp(-c/(4z^2))$ records the highest-weight-line projection of the Virasoro collision residue. The first correction beyond the leading power is

$$R_2 = -\frac{c}{4},$$

which is forced by the central collision coefficient $c/2$. This is a necessary normalisation check, not a sufficient non-formality theorem. The class M statement requires the completed Virasoro obstruction tower and the survival of the transferred wheel classes; it cannot be inferred from the scalar primary quotient alone.

6 Comparison with Heisenberg and affine Kac–Moody

The primary scalar factors provide a quick normalisation comparison between the three main families; they are not the full line-operator comparison.

Example 6.1 (Heisenberg: class G). The Heisenberg algebra $\mathcal{H}_k$ has OPE $J(z)J(w) \sim k/(z-w)^2$. The collision residue is $r(z) = k/z$, a simple pole. On the primary sector, $R(z) = z^k$: the R-matrix is a monomial. There is no exponential correction; $R_2 = 0$ and $S_3 = 0$. The tower terminates at degree 2: class G.

Example 6.2 (Affine Kac–Moody: class L). For $\widehat{\mathfrak{g}}_k$ with Casimir at level k, the collision residue is $r(z) = k\,\Omega_{\mathrm{tr}}/z$ in the trace convention and $r(z) = \Omega_{\mathrm{KZ}}/((k+h^\vee)z)$ in the Kazhdan–Lusztig convention, with $\Omega_{\mathrm{KZ}} = 2h^\vee\,\Omega_{\mathrm{tr}}$; the two conventions differ by a level-dependent rescaling of the Casimir, not by an identity in k. In the trace convention, $r\,|_{k=0} = 0$, while in the KZ convention $r\,|_{k=0} = \Omega_{\mathrm{KZ}}/(h^\vee z)$ remains finite and nonzero (the Lie bracket persists at abelian level). The R-matrix, in the KZ convention used throughout this table for uniformity across families, is $R(z) = z^{\Omega/(k+h^\vee)}$, a monomial in z with the Casimir eigenvalue as exponent. The cubic shadow $S_3 \neq 0$ (from the structure constants $f^{ab}{}_c$), so the tower extends to degree 3; the quartic and higher terms vanish by independent sum factorization: class L.

Example 6.3 (Virasoro: primary scalar factor). The scalar factor $R^{\mathrm{prim}}_{h,c}(z) = z^{2h}\exp(-c/(4z^2))$ has infinitely many nonzero coefficients. This reflects the irregular scalar highest-weight connection. The non-termination of the full Virasoro shadow tower is a separate completed-bar statement.

The comparison is summarized in the following table:

Family	scalar factor	S_3	$r_{\max}$	**Class**
Heisenberg $\mathcal{H}_k$	z^k	0	2	G
Affine $\widehat{\mathfrak{g}}_k$	$z^{\Omega/(k+h^\vee)}$	$\neq 0$	3	L
Virasoro Vir_c	$z^{2h}\exp(-c/(4z^2))$	2	∞	M

The progression from monomial (class G) through monomial with nontrivial exponent (class L) to essential singularity (class M) mirrors the increasing complexity of the OPE: quadratic, Lie-type, and non-quadratic respectively.

7 Level-prefix check: $k = 0$ and convention bridge

The Virasoro collision residue

$$r^{\mathrm{Vir}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}$$

carries the central charge only in the cubic central pole. At $c = 0$ the cubic pole vanishes, but the simple pole $2T/z$ remains in Vir_0 unless one further passes to the trivial quotient or to a vacuum scalar projection where T acts by zero. Thus $c = 0$ is a degenerate Virasoro point, not a trace-form vanishing check.

The *cubic + simple* pole profile is the correct one: d log-absorption sends the fourth-order OPE pole $c/2 \cdot (z-w)^{-4}$ to the third-order r-matrix pole $(c/2)/z^3$, and the second-order OPE pole $2T \cdot (z-w)^{-2}$ to the simple r-matrix pole $2T/z$. The simple OPE pole $\partial T/(z-w)$ is absorbed entirely and does not contribute. The common mistakes $r^{\mathrm{Vir}}(z) = (c/2)/z^4$ (fourth-order, un-absorbed) and $r^{\mathrm{Vir}}(z) = (c/2)/z^2$ (double pole, over-absorbed) are both excluded by the pole-order calculation.

Remark 7.1 (Trace-form vs λ-bracket convention bridge). Two conventions for the Virasoro r-matrix coexist in the framework:

(i) Trace-form (Vol I): $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$.

(ii) λ-bracket (Vol II): $\{T_\lambda T\} = (\lambda^3/6)\cdot c + 2\lambda T + T'$, equivalent to writing the cubic coefficient as $c/12$ in divided-power convention.

The bridge is $S_2^{\mathrm{Vol\,I}} = \kappa^{\mathrm{Vir}} = c/2 = 6S_2^{\lambda\text{-bracket}}$: the λ-bracket $c/12$ coefficient is related to the Vol I $c/2$ shadow invariant by the divided-power factor $3! = 6$. Bare $S_2 = c/12$ is the λ-bracket coefficient; the shadow invariant in Vol I notation is $S_2 = c/2$.

Remark 7.2 (Connection to κ and the shadow tower). The simple-pole coefficient $2T$ of $r^{\mathrm{Vir}}(z)$ averages (under av) to $2\cdot\kappa(\mathrm{Vir}_c) = c$ on the primary vacuum (since $\kappa^{\mathrm{Vir}} = c/2$, we have $2\kappa = c$). The cubic-pole coefficient $c/2$ is already a scalar: $\mathrm{av}(c/2) = c/2 = \kappa(\mathrm{Vir}_c)$. Both projections recover the modular characteristic, confirming the $E_1 \to$ shadow consistency: the cubic and simple poles project to the *same* κ scalar up to the factor 2 coming from the linear-in-T structure of the simple pole.

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